

Why Carbon Prices Alone May Not Deliver the Clean-Hydrogen Build-Out

Evidence on Implementation Risk under Transition Uncertainty

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Full research: github.com/SakeSaak/thesis_h2

Key messages

- Despite over USD 1.3 trillion in announced clean-hydrogen subsidies globally and a 422 GW project pipeline, only 7% of 2023-announced green hydrogen capacity reached Final Investment Decision [1]. The 2024–2026 wave of cancellations — ArcelorMittal (Bremen, Eisenhüttenstadt), BP (Teesside), and seven EU Hydrogen Bank auction winners (1.88 GW combined) — documents an implementation gap that price-only instruments have not closed.
- Cross-jurisdictional evidence indicates that policy effectiveness tracks the *number of economic frictions* an instrument addresses, not the per-unit subsidy magnitude. The EU Innovation Fund (>EUR 10 billion disbursed) produces a precise null on cancellation hazard across six estimators, while the China 14th Five-Year Plan and US Section 45Q — which combine financial support with coordinated demand commitments or output-linked credits — produce robust hazard reductions.
- Pre-FID offtake commitments are associated with an 11 to 13 percentage point reduction in cancellation hazard, robust across five matching estimators and an Oster (2019) sensitivity bound of $\delta_{\text{null}} = 20.23$. Demand certainty is the substantively under-appreciated mechanism through which policy effectiveness operates.
- The carbon-conditional cancellation-hazard intensity exhibits a structural break around 2020. Policy sensitivity is regime-dependent rather than stable; the empirical effect-magnitudes documented for the 2010–2024 window are not population-invariant.
- For ETS2 design and adjacent instruments: the evidence is consistent with the proposition that carbon-price signals alone are likely insufficient when demand certainty, infrastructure coordination, policy credibility, and implementation frictions are not simultaneously addressed.

1 Background: the green-hydrogen implementation gap

The European Union, the United States, the United Kingdom, and China have between 2018 and 2025 deployed an unprecedented combination of subsidies, capex grants, tax credits, and state-coordinated investment frameworks targeting the build-out of low-carbon hydrogen production

capacity. The aggregate announced subsidy envelope exceeds USD 1.3 trillion and the announced project pipeline has tripled to 422 GW between 2020 and 2024 [1].

Yet a substantively under-appreciated empirical regularity has emerged: the conversion rate from project announcement to Final Investment Decision (FID) has remained stubbornly low. [1] document that only 7% of 2023-announced green hydrogen capacity reached scheduled FID. The 2024–2026 period has crystallised this regularity into a sequence of high-profile cancellations:

- ArcelorMittal’s withdrawal from its Bremen and Eisenhüttenstadt DRI projects despite committed EU Innovation Fund subsidies of approximately EUR 1.3 billion (Q1 2025);
- BP’s exits from HyGreen Teesside and H2Teesside (Q4 2024);
- The September 2025 withdrawal of seven EU Hydrogen Bank auction winners, representing 1.88 GW of combined electrolysis capacity, before the security-deposit deadline.

The pattern is not confined to Europe. Air Products has cancelled three large US projects, and the post-2024 evaluation phase of US 45V three-pillars rules has been accompanied by visible project pipeline contraction. Industrial competitiveness debates in Brussels and member-state capitals have begun to converge on a shared diagnosis: subsidy availability, in itself, is no longer the binding constraint.

This brief presents evidence from a project-level cross-jurisdictional causal evaluation that helps identify which frictions *are* binding, and the design implications for ETS2 and adjacent instruments.

2 Implementation-risk framework: which frictions bind?

A clean-hydrogen FID is a staged, irreversible decision under transition uncertainty. The sponsor’s expected net payoff depends on a set of economic frictions, each of which may or may not bind in a particular policy environment. Seven candidate frictions are catalogued in the underlying research [3]. For policy-design purposes, five are decision-relevant:

- **F1 — Production cost gap.** Per-unit production cost above the avoided-emission value at prevailing carbon prices. Addressed by output credits (US 45V/45Q) and capex grants (EU Innovation Fund).
- **F2 — Infrastructure coordination.** Pipeline networks, port facilities, electricity grid capacity for electrolysis. Addressed by cluster tenders (UK Track-1) and state-coordinated mandates (China 14th Five-Year Plan).
- **F3 — Policy credibility.** Belief that the policy environment will persist over the multi-decade investment horizon. Addressed (or undermined) by legislative permanence, election cycles, and treaty consistency.
- **F4 — Revenue uncertainty (demand certainty).** Whether expected offtake at known prices is contractually secured pre-FID. Addressed by offtake-commitment mandates, contracts-for-difference, hydrogen auctions with price-floor guarantees.
- **F5 — Implementation friction.** Permitting timelines, supply-chain bottlenecks, skilled-labour availability, electrolyser-stack delivery schedules. Addressed by streamlined permitting and dedicated supply-chain interventions.

The proposition advanced by the underlying empirical research is that *policy effectiveness is determined by the joint binding-friction structure*, not by the magnitude of any single instrument. An instrument that addresses a non-binding friction (such as F1 in the contemporary capital-abundant clean-technology environment) will produce a precise null in cancellation-hazard reduction even when the subsidy magnitude is large. An instrument that addresses a binding friction (such as F4 demand certainty) can produce a substantial effect at modest financial cost.

3 Regime dependence under transition uncertainty

A second methodological feature of the research is that the binding-friction structure itself *evolves with the policy regime*. The carbon-conditional cancellation-hazard differential between Blue (CCS-equipped) and Green (electrolysis-based) hydrogen projects, denoted $\beta_{\text{int}}(t)$, is documented to intensify from approximately -0.46 in 2010 to -1.49 in 2024, with a discernible structural break around 2020 coinciding with the European Green Deal regime boundary [2, 3].

This finding has two consequences for ETS2 design. First, policy-effect magnitudes estimated on pre-2020 data systematically miss the post-2020 regime: the binding-friction structure has shifted, and effect-magnitudes have shifted with it. Second, the regime-dependence implies that ETS2 effectiveness will not be invariant to the contemporaneous structure of complementary instruments. The carbon-price signal interacts with the other four frictions; its marginal effectiveness depends on whether F2 (coordination), F4 (demand certainty), and F5 (implementation) are simultaneously addressed.

4 Empirical evidence: three findings

The underlying research — a 159-page bridging thesis with four companion papers and a 48-script robustness suite, all archived under DOI [10.5281/zenodo.20359771](https://doi.org/10.5281/zenodo.20359771) — documents three principal findings relevant for ETS2 and clean-hydrogen policy.

Finding 1 — Pre-FID offtake commitments mitigate F4 demand uncertainty

Pre-FID offtake commitments are associated with an 11 to 13 percentage point reduction in cancellation hazard. The effect is robust across five independent matching strategies (inverse-probability-weighted regression adjustment, ordinary-least-squares regression-adjusted, regression-adjusted matching, doubly-robust, and naive inverse-probability weighting). An Oster (2019) selection-on-unobservables sensitivity analysis returns $\delta_{\text{null}} = 20.23$, indicating that an implausibly large omitted-variable bias would be required to overturn the finding. The effect concentrates in high-volatility sectors (power and heat, transport) and is small or null in low-volatility sectors (chemical, refinery), consistent with a real-options interpretation in which demand certainty resolves multiple uncertainty channels simultaneously.

Finding 2 — Carrot-policy effectiveness ranks by friction-count, not subsidy magnitude

Cross-jurisdictional evaluation via modern difference-in-differences estimators (Sun-Abraham 2021; Borusyak-Jaravel-Spiess 2024; Callaway-Sant’Anna 2021) produces a consistent ranking:

China 14th Five-Year Plan > US Section 45Q > UK Track-1 >> EU Innovation Fund

The ranking tracks the number of binding frictions each instrument simultaneously addresses, not the per-unit monetary value of the subsidy. The EU Innovation Fund informative null — a precise null across six estimators despite EUR 10 billion+ disbursed — is interpreted as substantive evidence that the production cost gap (F1) is not the binding friction in the contemporary capital-abundant clean-technology environment. Rambachan-Roth (2023) honest-DiD sensitivity bounds reveal heterogeneous identification strength across the four instruments: only China’s state-coordinated mandate survives the strictest robustness checks ($M^* = 1.5$), while US, EU, and UK estimates are sensitivity-bounded.

Finding 3 — Policy sensitivity is regime-dependent

The carbon-conditional Blue/Green cancellation-hazard differential $\beta_{\text{int}}(t)$, estimated through a score-driven Generalized Autoregressive Score (GAS) state-space specification, intensifies across the 2010–2024 window with a structural break around 2020. Constant-parameter and parameter-driven random-walk specifications systematically miss this break. The Diebold-Mariano-Harvey-Leybourne-Newbold out-of-sample forecast comparison and the Hansen-Lunde-Nason Model Confidence Set both select the score-driven specification as the singleton.

5 Implications for ETS2 design

The evidence cumulates into a coherent design observation. **Carbon-price signals operate on F1 (production cost gap), but F1 does not currently appear to be the binding friction.** The EU Innovation Fund informative null is the cleanest empirical demonstration: a large financial transfer that directly addresses F1 produces no detectable cancellation-hazard reduction in the contemporary environment. The bindings are elsewhere — in F2 (infrastructure coordination), F3 (policy credibility), F4 (demand certainty), and F5 (implementation friction).

For ETS2, this evidence is consistent with the following design proposition:

The cancellation-hazard reduction achievable through carbon-price strengthening alone is bounded by the magnitude of the non-binding friction (F1) it addresses. To recover the unbinding-friction-magnitudes, ETS2 design should be evaluated jointly with complementary instruments addressing demand certainty (F4), infrastructure coordination (F2), and policy credibility (F3) — not as an independent lever.

Four concrete design corollaries follow from the empirical evidence, presented here as evidence-positioned observations rather than normative recommendations:

1. **ETS2 effectiveness will depend on parallel demand-certainty instruments.** The offtake-commitment finding suggests that an ETS2-induced carbon-price increase will translate into accelerated FID only to the extent that downstream offtake at predictable prices is contractually secured. EU-level mechanisms creating demand certainty — the European Hydrogen Bank, contracts-for-difference, or sectoral offtake-quota frameworks — are the natural complements.
2. **Combining instruments may yield super-additive effects.** The friction-count ranking implies that pairs of instruments addressing distinct frictions produce larger effects than the sum of their individual effects. Counterfactual scenario analysis under explicit structural-invariance assumptions estimates that an EU-wide hard-link between Innovation Fund

eligibility and demonstrable offtake commitments could deliver +83 additional FIDs and +858 kt/y of additional green hydrogen capacity over a three-year horizon [3].

3. **Policy credibility (F3) is a separate decision-variable.** The regime-dependence finding suggests that signal-strengthening (ETS2 expansion) without credibility-strengthening (legislative permanence, treaty consistency, election-cycle robustness) is likely to produce smaller effects than credibility-strengthening alone. Policy permanence is a separable design parameter, not a by-product of signal magnitude.
4. **Per-instrument magnitudes are not population-invariant.** The Rambachan-Roth honest-DiD sensitivity bounds across instruments imply that estimated effect-magnitudes for ETS2 derived from any one historical window should not be extrapolated to future-window effectiveness without explicit regime-invariance assumptions.

6 Methodological caveats

The empirical evidence presented here is L3.a quasi-causal (multi-estimator convergence under modern difference-in-differences parallel-trends or conditional-independence assumptions) and L3.b partial-identified (Rambachan-Roth and Oster sensitivity bounds). The channel-attribution to specific frictions (F1–F5) is interpretive rather than directly identified: three formal identification tests [3] document that the channels are not empirically structurally separable in the available observational sample. The counterfactual scenarios are L4 extrapolations conditional on structural-invariance over the projection window and on the absence of general-equilibrium feedback. They should be read as decision-relevant illustrations of per-mechanism leverage rather than as point predictions of realised outcomes.

The 2024–2026 cancellation events serve as out-of-sample corroboration of the implementation-risk framework, but the framework itself is interpretive: it organises the observed evidence rather than claiming to predict each individual cancellation event.

7 Companion research and references

This brief synthesises evidence from a 159-page bridging thesis and four standalone companion papers. The complete research portfolio, including raw analysis scripts and a 48-component robustness suite, is publicly archived at DOI 10.5281/zenodo.20359771 under CC-BY 4.0 (written work) and MIT (code) licenses.

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